2019 EXAMINATIONS

Part I (First Year)

SCHOOL OF COMPUTING AND COMMUNICATIONS

SCC.110 Software Development (1 hour & 30 minutes)

- **Answer any <u>THREE</u> out of the four questions.**
- **Use a <u>separate</u> answer book for <u>each</u> question.**

QUESTION 1 (25 Marks)

1.a The following question concerns the **C language**.

i. The C programming language supports the use of functions. Provide 3 benefits that arise from structuring your programs using functions.

[3 Marks]

ii. Functions can return a range of different types. Provide one example of a valid return type *and* one example of an invalid return type.

[2 Marks]

1.b For each of the following statements about functions in C, briefly state whether they are true or false and provide a brief explanation or clarification as appropriate:

i. Functions can return multiple values.
ii. C passes parameters by reference.
iii. Inside your `main()` function you can always access variables declared inside any other functions in the program.
iv. If you do not declare a return type for a function it will default to void.
v. Both `int main()` and `int main(int argc, char *argv[])` are valid headers for the main function.

[5 Marks]

1.c You have been asked to help a local cycle shop with a problem with their software. When you investigate you discover that the functions needed to support the program seem to be missing. Read the program and documentation below and then answer the following questions.

Question 1 continues on next page…

Question 1 continued.

```
int main ()
{
  int leg, wheelSize, inputTeeth, outputTeeth;
  float ratio;

  printf ("Enter the child's leg measurement in cm.\n");
  scanf ("%d",&leg);
  wheelSize = calculate_wheel_size (leg);
  printf ("The correct wheel size is %d.\n", wheelSize);

  printf ("Enter the number of teeth on the input and
output cog.\n");
  scanf ("%d,%d",&inputTeeth, &outputTeeth);
  ratio = calculate_gear_ratio(inputTeeth, outputTeeth);
  printf ("The gear ratio is %f.\n", ratio);

  shift_up(&inputTeeth, &outputTeeth);
  ratio = calculate_gear_ratio(inputTeeth, outputTeeth);
  printf ("The gear ratio is %f.\n", ratio);
 }
```

Documentation

Leg measurement (cm)	**Wheel Size**
<24	Too short (return -1)
24	10
31	12
38	14
>48	Too tall (return -1)

Gear ratios are calculated by dividing the number of teeth on the drive gear (input teeth) by the number of teeth on the driven gear (output teeth).

Shifting up a gear means doubling the number of teeth on the drive gear and halving the number of teeth on the driven gear.

Question 1 continues on next page…

Question 1 continued.

i. Write the function `calculate_wheel_size()`. You should make sure to include a suitable function header and return the correct value.

[3 Marks]

ii. Write the function `calculate_gear_ratio()`. You should make sure to include a suitable function header and return the correct value.

[3 Marks]

iii. Write the function `shift_up()`. You should make sure to include a suitable function header and return the correct values.

[3 Marks]

1.d Professional software developers apply a set of style guidelines when writing their code in order to make their code easier to maintain, read and share with others. The following code snippet has been written by a poor programmer:

```
#include <stdio.h>

int main() {int i, x,y,z = 0;

  int test[] = {1,5,7,2,9,11}; x = test[0];y = test[0];

  for (i=0; i<6;i++) {
    if (test[i]<x)
x = test[i];
    if (test[i]>y) y = test[i];
    z+=test[i];
                        }
  printf ("%d,%d,%d\n",x,y,z);}
```

i. Find and list four stylistic faults in this code.

[4 marks]

ii. What is the code doing ?

[2 marks]

[Total 25 Marks]

QUESTION 2 (25 Marks)

Imagine you have been contracted to develop a piece of Java software for the ICESAT company. ICESAT place satellites in earth orbit to monitor the condition of ice flows and icebergs across the world.

You have been tasked with creating a piece of software to store information on a number of icebergs around the world, in order to provide a warning system to any ships at sea if they sail too close to the known location of an iceberg. You have been asked to use the Java programming language.

2.a Write a Java class to represent an **Iceberg**. This class should contain instance variables to hold the following pieces of information. Be careful to choose the best types to hold this information, and ensure your class is properly encapsulated.

- A number that identifies the iceberg (you may assume this is a whole number).
- The size of the iceberg recorded as its maximum diameter in metres.
- The location of the centre of the iceberg, using simple x and y coordinates.

[4 Marks]

2.b Write an appropriate constructor for your Iceberg class.

[3 Marks]

2.c You have also been asked to store information on all ships registered with ICESAT. Write another class called **Ship**. Each ship has a name, length (in metres) and a location also represented using simple x and y co-ordinates. Create an appropriate constructor for this class.

[3 Marks]

2.d Write one further class that represents the Earth. This should contain two arrays as instance variables – an array of all the icebergs on the Earth, and an array of all the known ships. For the purpose of this exercise, you may assume that there will only be a maximum of three ships and three icebergs.

[4 Marks]

Question 2 continues on next page…

Question 2 continued.

2.e Show how you could create a constructor for your Earth class (with no parameters) that populates its arrays with instances of your Iceberg and Ship classes with the following values:

Iceberg id	diameter	x	y
1	33.1	101010.56	4321.321
42	298.7	0.0	0.0
666	6989	6789.98	9876.67

Ship Name	length	x	y
HMS Tiddler	200	689.9	41.1
HMS Godzilla	300	6780.0	9700.0
HMS Behemoth	1800	921.3	400.0

[3 Marks]

2.f Ships are defined as being in danger of colliding with an iceberg if they get too close. An engineer at ICESAT decides that if a ship is less that its length plus the diameter of an iceberg away from that iceberg, then it is in danger of collision. **Paying particular attention to the principles of encapsulation**, write one or more methods that print the list registered ships are in danger of colliding with an iceberg. State clearly in which class you would add these method(s).

Remember that the distance between two points can be determine using Pythagoras' theorem:

$$distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

You may also find the following standard Java methods useful:

`double` **`Math.pow`**`(double a, double b)`
Returns the value of the first argument raised to the power of the second argument.

`double` **`Math.sqrt`**`(double a)`
Returns the correctly rounded positive square root of a double value.

[8 Marks]

[Total 25 Marks]

Question 3 (25 Marks)

3.a

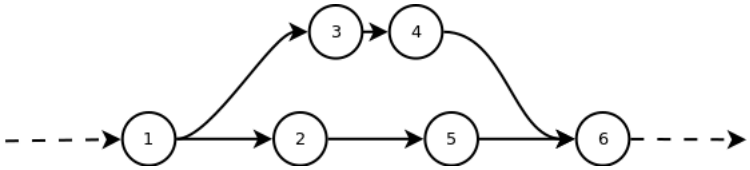
i. There are two common models used for version control software. State both and explain their key differences.

[5 Marks]

ii. Git internally uses one of these two models. Explain why git *can* be used for either approach and name an example system using git this way.

[4 Marks]

3.b The diagram below is a graph of a small number of commits (the numbered circles) in a git history. With reference to the numbered commits, state what operations were performed at each stage. Note that time flows from left to right in the diagram.



[8 Marks]

3.c Explain how a *merge conflict* may occur when multiple people are using a git repository. Be specific in your answer.

[3 Marks]

3.d Consider a program written in Java, which should take a string as input and outputs a new string rearranged to put all the characters in alphabetical order, as illustrated below:

```
inOrder("helloworld") -> "dehllloorw"
```

Provide at least **five** equivalence partitions and their test cases to test this program.

[5 Marks]

[Total 25 Marks]

QUESTION 4 (25 Marks)

4.a Consider the following C program.

```
1  #include <stdio.h>
2  #include <string.h>
3
4  char * cs(char *from, char *to, int sChar, int dChar)
5  {
6     char *result = malloc(strlen(to) + 2);
7     int i;
8
9     for(i = 0; i < dChar; i++)
10       result[i] = to[i];
11
12    result[i] = from[sChar];
13
14    for(i = dChar + 1; i < strlen(to) + 1; i++)
15       result[i] = to[i-1];
16
17    result[i] = 0;
18
19    return result;
20 }
21
22 int main()
23 {
24    char *s = "pound";
25    char *d = "or";
26
27    printf("%s %s\n\n", s, d);
28
29    char *result = cs(s, d, 2, 1);
30    printf("%s\n", result);
31 }
32
```

i. Describe as precisely as you can the meaning of the code on the following lines:

- Line 4
- Line 6
- Line 9
- Line 27
- Line 29

[5 marks]

Question 4 continues on next page...

Question 4 continued.

ii. Why are there so few curly {} brackets in this program?

[1 mark]

iii. 'Dry run' the program starting at line 22. Describe the logical flow of execution of the program using the line numbers to help.

[5 marks]

iv. What is the purpose of the program? What does the program print as a result?

[2 marks]

Question 4 continues on next page…

Question 4 continued.

4.b Consider the following Java program:

```
1  public class Something
2  {
3      public void doSomething(int[] x)
4      {
5          int m=0;
6
7          for (int i=0; i < x.length; i++)
8          {
9              if (x[i] < 0)
10                 x[i] = 0;
11
12             if (m < x[i])
13                 m = x[i];
14         }
15
16         for (int i=m; i>0; i--)
17         {
18             String s = "";
19             for (int j=0; j < x.length; j++)
20             {
21                 if (x[j] >= i)
22                     s = s + "*";
23                 else
24                     s = s + " ";
25             }
26
27             System.out.println(s);
28         }
29     }
30
31     public static void main (String[] args)
32     {
33         Something s1 = new Something();
34
35         int[] a = {3,6,-1,0,4};
36
37         s1.doSomething(a);
38     }
39 }
```

i) Describe as precisely as you can the meaning of the code on the following lines:

- Line 1
- Line 3
- Line 7
- Line 33
- Line 37

[5 Marks]

ii) Analyse the above program. Describe as precisely and concisely as you can what the program does. Write down what the program would print to the screen.

[7 Marks]

[Total 25 Marks]

---End of Paper---